

Invoice Generator

Product & Technical Documentation

GST-ready invoicing, multi-currency billing, payment tracking, PDF documents and email delivery, built on Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, Drizzle ORM and PostgreSQL.

Item	Value
Application	Invoice Generator (version 0.1.0)
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1. Overview

Invoice Generator is a web application for small and medium businesses, freelancers and agencies that need to issue professional invoices, record payments against them and understand their receivables. It is built around Indian GST rules (CGST/SGST for intra-state supplies, IGST for inter-state supplies, GSTIN capture, HSN/SAC codes and financial-year based numbering) while also supporting billing in foreign currencies with rates frozen on each invoice.

Each registered account is a separate **business workspace**. All clients, invoices, payments and settings belong to that workspace and are never visible to another. Reporting is normalised to Indian Rupees (INR) so that a business billing in USD, EUR, GBP or AED still sees one consistent picture of revenue and outstanding amounts.

1.1 What the application does

- **Client management:** B2B and B2C clients with contact details, GSTIN, address, state code, preferred currency and notes. Clients can be archived and restored.
- **Invoicing:** multi-line invoices with quantity, unit, unit price, per-line tax rate and HSN/SAC code; invoice-level percentage or fixed discounts; automatic GST split; draft and issued states; sequential numbers that reset every financial year.
- **Multi-currency:** invoices in INR, USD, EUR, GBP or AED with a live or manual exchange rate frozen on the invoice.
- **Payments:** full or partial payments by bank transfer, UPI, card, cash, cheque or other, each producing a numbered receipt; invoice status updates automatically.
- **Documents:** A4 PDF invoices (titled TAX INVOICE for GST B2B supplies) and payment receipts, previewable in the browser or downloadable.
- **Email:** send an invoice with the PDF attached to the client through Resend, with a full delivery log per invoice.
- **Dashboard and reports:** revenue, collected, pending and overdue totals, monthly invoiced-vs-collected charts, status breakdowns, top clients and per-currency analysis for any date range.
- **Transaction history:** one merged, filterable timeline of invoices and payments.
- **Business settings:** company profile, address, GST configuration, invoice prefix, default currency, payment terms, bank details and default notes/terms.
- **Appearance:** seven professional colour themes and light, dark or system mode, persisted per browser.

1.2 Who uses it

Role	Typical use
Business owner / accountant	Issues invoices, records payments, sends documents to clients, reviews reports at month and year end.
Administrator	Sets up the business profile, GST details, invoice numbering prefix and email delivery.
Developer	Extends the application; the stack and architecture sections of this document are written for them.

2. Features in detail

2.1 Authentication and workspaces

- Registration creates a business and its first user in one step and signs the user in. New users land on Business settings with a welcome notice.
- Login accepts email and password. A failed attempt returns a generic message so that valid emails cannot be discovered.
- Sessions last 30 days and are stored in the database; only a hash of the session token is kept server-side (details in section 8).
- Every dashboard page requires a signed-in user. Signed-in users visiting the login or register page are sent to the dashboard.
- One user belongs to exactly one business. Team invites, roles and password reset are not implemented yet.

2.2 Dashboard

- Four overview cards: **Total Revenue** (issued invoices in INR, with month-over-month change), **Paid Amount** (payments received), **Pending Amount** (balance of pending and partially paid invoices) and **Overdue Amount**.
- **Revenue overview**: a 12-month invoiced-vs-collected bar chart with hover details.
- **Invoice status**: counts and percentages for paid, pending, overdue and draft.
- **Recent invoices** (latest 6) and **recent payments** (latest 5) with links to the full lists.
- Quick actions: Add client and New invoice. The sidebar shows a live badge with the number of unpaid invoices.

2.3 Clients

- List with search, B2B/B2C filter and an archived toggle. Columns: client, contact, GSTIN, number of invoices, amount invoiced (INR) and outstanding.
- Client record: type, name, contact name, email, phone, GSTIN, address lines, city, state and GST state code, postal code, country (default India), preferred currency and notes.
- Client detail page shows the full invoice history with issued, invoiced and outstanding totals, plus Edit, Archive/Restore and New invoice (pre-selecting the client).
- Archiving is a soft delete: the client is hidden from pickers and search but history is preserved.

2.4 Invoices

Creating an invoice

- Choose a client. The invoice type (B2B/B2C), place of supply and currency are pre-filled from the client and can be changed.
- Set issue date and due date (the due date defaults from the business payment terms and may not be before the issue date).
- Add one or more line items: description, HSN/SAC, quantity (3 decimals), unit, unit price, tax rate (default 18% when GST is enabled, 0% otherwise).
- Optionally apply an invoice-level discount, either a percentage or a fixed amount, which is allocated proportionally across lines before tax.

- For non-INR invoices the current exchange rate is fetched automatically and can be refreshed, or entered manually.
- Totals (subtotal, discount, taxable amount, CGST/SGST or IGST, total and INR equivalent) update live as you type.
- Save as **draft** or **issue** immediately. A sequential invoice number is assigned either way and is never reused.

Invoice detail and actions

- Header: number, status, B2B/B2C, issue and due dates, financial year and sent date.
- Cards for Bill to, Tax & currency (rate and source, GST mode, place of supply, GSTINs), Balance (total, paid, balance due, INR total), the items table and notes/terms.
- Actions: Preview PDF, Download PDF, Send email, Record payment, Issue (drafts), Edit (while editable), Delete draft or Cancel invoice.
- Payments received against the invoice are listed with receipt download and delete; the email history lists every send attempt with provider and outcome.

Invoice list

- Paginated (25 per page), searchable by number or client, filterable by status, client and issue-date range.

Invoice statuses

Status	Meaning
Draft	Saved but not issued. Fully editable; can be deleted. Not counted in revenue.
Pending	Issued and awaiting payment, not yet past the due date.
Partially paid	At least one payment recorded, balance still outstanding.
Paid	Balance due is zero. The paid-at timestamp is recorded.
Overdue	Pending or partially paid and the due date has passed. Updated automatically whenever lists, the dashboard or reports load.
Cancelled	Voided before any payment. Cannot be edited or paid. Retains its number for the audit trail.

An invoice can be edited only while it is draft, pending or overdue **and** nothing has been paid against it. Editing keeps the invoice number and recalculates all totals. Invoices with payments must be cancelled or settled rather than changed.

2.5 Payments and receipts

- Record payment from the invoice page: amount (defaults to the full balance), date, method, reference such as a UTR or cheque number, and notes.
- Payments are accepted only against pending, partially paid or overdue invoices, and never for more than the outstanding balance. Partial payments are allowed.
- Each payment receives a receipt number in the form RCPT/<financial year>/<sequence> and a PDF receipt can be downloaded.
- The INR value of every payment is frozen using the invoice's exchange rate, so collections are comparable across currencies.
- Deleting a payment recomputes the invoice's paid amount, balance and status.

- The Payments page lists all payments with search, method and date filters, and shows the total received in INR.

2.6 PDF documents

Document	Contents
Invoice PDF	Title TAX INVOICE (GST applied, B2B) or INVOICE. Business block with GSTIN and PAN, bill-to block with client GSTIN, issue/due dates, type, currency, place of supply and exchange rate, items table (HSN/SAC shown when GST applies), subtotal, discount, taxable amount, CGST+SGST or IGST or plain tax, total, amount paid, balance due, INR equivalent, payment details, notes, terms.
Payment receipt PDF	Title PAYMENT RECEIPT with the receipt number, business and client details, payment date, method and reference, the invoice it settles, amount received, remaining balance and INR equivalent.

Both documents are A4, generated on the server, and can be opened inline in the browser or downloaded. File names follow the document number, for example INV-2026-27-0001.pdf.

2.7 Email delivery

- Send an invoice to the client's email (or any override address) with an optional personal message. The PDF is attached and a branded HTML email with amount due, total, dates, a View invoice button and payment details is sent.
- Replies go to the business billing email. Subject line: *Invoice <number> from <business>*.
- Every attempt is logged per invoice with recipient, time, provider, message id and any error.
- Sending an issued draft promotes it to pending and records the sent date.
- Without an email API key the application runs in **console mode**: the email is printed to the server log instead of being delivered, and the interface says so clearly.

2.8 Reports

- Date range picker with quick ranges for the current and previous financial year and the last 30 days. Defaults to the current Indian financial year (1 April to 31 March).
- Summary: invoiced amount and count, collected amount and collection rate, outstanding and overdue amounts, tax billed and average invoice value.
- Monthly revenue chart (invoiced vs collected), top 10 clients with outstanding balances, invoice statistics by status, and a per-currency breakdown with original and INR totals.
- All figures are in INR using the rates frozen on each invoice.

2.9 Transaction history and Records

History is a single timeline of invoices (by issue date) and payments (by payment date), filterable by text, kind, client and date range. Payments are shown in green with a plus sign and reference the invoice they settle.

Records, available from the sidebar and as an All records card on the dashboard, lists every client, invoice and payment in the workspace in three tabs with counts, each row linking to its detail page and a Manage button opening the full list.

2.10 Business settings

Section	Fields
Business profile	Name, legal name, billing email (used as reply-to), phone, website, PAN
Address	Address lines, city, state (dropdown of GST state codes when the country is India), postal code, country
Tax (GST)	GST enabled switch and GSTIN. When enabled, the GSTIN and business state are mandatory.
Invoice defaults	Invoice prefix with a live preview, default currency, payment terms in days, payment/bank details printed on invoices, default notes and default terms

Changes apply to new invoices only; existing invoices keep the values that were in effect when they were created.

2.11 Search and navigation

- Global search in the header: choose Invoices or Clients, type to see suggestions, press Enter to open the first match.
- Collapsible sidebar with Dashboard, Clients, Invoices, Payments, History, Records, Reports and Settings. On mobile it becomes a slide-in panel.
- Breadcrumb showing workspace and current page; account menu with Business settings, Theme, Appearance and Log out.

2.12 Appearance

Seven accent themes (Graphite, Navy, Ocean, Indigo, Teal, Forest, Slate) and three modes (Light, Dark, System) are available from the account menu. The choice is stored in the browser and applied before the page paints, so there is no flash of the default theme.

3. How it works: core business rules

3.1 Document numbering and financial year

- Invoice numbers: **<PREFIX>/<FY>/<NNNN>**, for example INV/2026-27/0001. The prefix is configurable (1 to 10 upper-case letters or digits, default INV).
- Receipt numbers: **RCPT/<FY>/<NNNN>**.
- The financial year follows the Indian convention, 1 April to 31 March, labelled YYYY-YY. The invoice FY comes from the issue date and the receipt FY from the payment date.
- Sequences reset to 0001 at the start of each financial year and are tracked per business, per document kind, per financial year.
- Numbers are allocated atomically inside the same database transaction that saves the document, using an insert-or-increment on the counters table, so two users saving at once can never receive the same number. A unique index on (business, invoice number) is a second safeguard.
- Numbers are consumed by drafts too and are never reused, even if the draft is deleted. This keeps the sequence audit-friendly.

3.2 Tax and GST calculation

One pure calculation function runs in the browser for live totals and on the server for the saved values, so both always agree. All arithmetic uses decimal numbers (never floating point) with half-up rounding to 2 decimals.

Step	Rule
1. Line subtotal	quantity x unit price, rounded to 2 decimals
2. Invoice discount	Percent: subtotal x rate / 100. Fixed: the entered amount. Clamped between 0 and the subtotal.
3. Discount allocation	Spread across lines in proportion to each line subtotal; any rounding remainder goes to the last non-zero line.
4. Taxable amount	line subtotal minus the allocated discount
5. Tax, GST intra-state	CGST = SGST = taxable x rate / 100 / 2 (each half rounded first). Applies when the place of supply equals the business state.
5. Tax, GST inter-state	IGST = taxable x rate / 100. Applies when the place of supply differs from the business state, or when no state is known.
5. Tax, GST disabled	Plain tax = taxable x rate / 100, shown as a single Tax line.
6. Total	(subtotal minus discount) plus the sum of line taxes
7. INR total	total x exchange rate, rounded to 2 decimals and frozen on the invoice

- Tax rates are set per line (0 to 100, two decimals) rather than from a fixed slab list, so 0%, 5%, 12%, 18%, 28% or any other rate can be used.
- The place of supply defaults to the client's state code, then the business state code, and can be overridden on the invoice.
- GSTINs, the place of supply, the GST mode and the inter-state decision are **snapshotted** on the invoice so later changes to the business or client never alter historical documents.
- GSTINs are validated against the official 15-character pattern and upper-cased. The state list contains all 38 GST state codes plus Other Territory (97) and Outside India / Export (96).

- If GST is enabled but the business state is missing, the invoice form shows a warning because every supply would be treated as inter-state.

3.3 Currency and exchange rates

Code	Currency	Symbol
INR	Indian Rupee (base and reporting currency)	₹
USD	US Dollar	\$
EUR	Euro	€
GBP	British Pound	£
AED	UAE Dirham	AED

- Each invoice has its own currency, defaulting to the client's currency and then the business default.
- Rates are fetched from the free open.er-api.com service (INR base, no API key) with an 8-second timeout, inverted to INR per unit, stored to 8 decimals and cached for 12 hours (configurable).
- If a refresh fails the last cached rate is used and flagged as stale; if nothing is cached the user is asked to enter the rate manually.
- Rate source is recorded as base (INR), api or manual. The rate is frozen on the invoice and reused for every payment against it.

3.4 Payment application and status transitions

Recording a payment locks the invoice row, validates the status and amount, allocates the receipt number, stores the payment with its INR value and then recomputes the invoice:

- amount paid = sum of all payments; balance due = total minus amount paid (never below zero).
- balance zero: **paid** (paid-at set). Some paid, balance remaining: **partially paid**, or **overdue** if past due. Nothing paid: **pending**, or **overdue** if past due.
- Draft and cancelled invoices are never changed by this process.
- Deleting a payment runs the same recomputation, so an invoice can move back from paid to partially paid.

3.5 Multi-tenancy and data isolation

Every domain table carries the owning business id, and every query filters on it. A user can only ever read or change data belonging to their own business. The only shared table is the exchange-rate cache, which contains no business data.

4. Technology stack

Layer	Technology	Version	Role
Framework	Next.js (App Router, Turbopack)	16.3.8	Routing, server components, server actions, route handlers
UI library	React / React DOM	19.2.8	Component model
Language	TypeScript	5.9	Strict typing across app, schema and validation
Styling	Tailwind CSS	4.x	Utility CSS via @tailwindcss/postcss; no tailwind.config
Components	shadcn/ui (base-nova style) on @base-ui/react	4.21 / 1.8	Accessible primitives: dialog, dropdown, select, sidebar, table, calendar, etc.
Icons	lucide-react	1.50	Icon set
Forms	react-hook-form + @hookform/resolvers	7.89 / 5.9	Form state, field arrays for invoice lines
Validation	zod	4.6	Shared schemas for client and server validation
Money	decimal.js	10.6	Exact decimal arithmetic, 28-digit precision, half-up rounding
Database	PostgreSQL	16 / 17	Primary data store
ORM	Drizzle ORM + drizzle-kit	0.45 / 0.31	Typed schema, queries, SQL migrations
DB driver	postgres (postgres.js)	3.4	Connection pool; prepared statements disabled for pooler compatibility
PDF	@react-pdf/renderer	4.9	Server-side invoice and receipt PDFs
Email	Resend + @react-email/components	6.32 / 1.0	Transactional email with React templates
Dates	date-fns, react-day-picker	4.4 / 10.0	Calendar date picker
Notifications	sonner	2.0	Toast messages
Fonts	Geist Sans, Geist Mono		Loaded through next/font
Tooling	ESLint 9 (eslint-config-next), Node.js		Linting; Node 20+ (Node 25 in development)

4.1 Architecture at a glance

```
Browser (React 19 client components, Tailwind, shadcn/ui)
| server actions (mutations)      | GET navigation (server components)
v                                v
Next.js 16 App Router  -- proxy.ts optimistic auth gate (session cookie)
| lib/actions/*  -> validate (zod) -> lib/data/* (business logic)
| app/api/*      -> PDF route handlers (@react-pdf/renderer)
v
Drizzle ORM (lib/db.ts, postgres.js) ---> PostgreSQL (10 tables, 7 enums)
|
+--> Resend (email)          +--> open.er-api.com (exchange rates)
```

- **Server Components** fetch data directly through lib/data and render pages; **Client Components** are used only where interactivity is needed (forms, dialogs, menus, charts hover).

- **Server Actions** in `lib/actions` handle every mutation. Each returns a uniform result object: ok with data, or ok false with an error and optional field errors.
- **Validation happens twice**: in the browser via `react-hook-form` with the `zod` resolver, and again on the server with the same `zod` schema.
- **Route Handlers** under `app/api` serve PDF bytes with inline or attachment disposition and private no-store caching.
- **proxy.ts** (the Next.js 16 replacement for middleware) redirects unauthenticated requests away from the dashboard and signed-in users away from the auth pages. The authoritative check is the database-backed session lookup on every protected page.

4.2 Project structure

```

app/
  login/, register/      authentication pages
  dashboard/             layout, loading, error + clients/, invoices/,
                        payments/, history/, records/, reports/, settings/
  api/invoices/[id]/pdf  invoice PDF route handler
  api/payments/[id]/receipt receipt PDF route handler
components/
  auth/, charts/, clients/, dashboard/, invoices/, payments/,
  settings/, shared/, theme/, ui/ (shadcn)
db/schema/              Drizzle table definitions (one file per domain)
db/migrations/          generated SQL migrations
lib/
  actions/              server actions (auth, clients, invoices, payments, settings, search)
h)
  data/                 queries and business logic per domain
  auth/                 password hashing, sessions, current user
  pdf/                  invoice and receipt documents, styles
  email/                provider abstraction, send-invoice, templates
  validation/           zod schemas
  calculations.ts, money.ts, currency.ts, document-number.ts,
  fiscal-year.ts, india-states.ts, format.ts, theme.ts, db.ts
proxy.ts                auth gate
drizzle.config.ts       Drizzle Kit configuration

```

5. Data model

The schema is defined in TypeScript with Drizzle and generated into SQL migrations. Column names are snake_case in the database and camelCase in code. All money columns use PostgreSQL numeric (14,2); exchange rates use numeric (18,8); quantities use numeric (12,3). Floating point is never used for money.

5.1 Tables

Table	Purpose	Key columns
businesses	The tenant / workspace	name, legal_name, email, phone, website, address fields, state_code, country, gst_enabled, gstin, pan, default_currency, invoice_prefix, payment_terms_days, bank_details, invoice_notes, invoice_terms
users	Login accounts (one business each)	business_id, name, email (unique, lower-case), password_hash
sessions	Server-side login sessions	id (SHA-256 of token), user_id, expires_at
clients	Customers	business_id, type (b2b/b2c), name, contact_name, email, phone, gstin, address fields, state_code, currency, notes, archived_at
invoices	Invoice header with frozen snapshots	business_id, client_id, invoice_number (unique per business), financial_year, invoice_type, status, issue_date, due_date, currency, exchange_rate, exchange_rate_source, gst_applied, business_gstin, client_gstin, place_of_supply(_code), is_inter_state, subtotal, discount_type/value/amount, taxable_amount, cgst/sgst/igst/tax_amount, total, amount_paid, balance_due, total_inr, notes, terms, sent_at, paid_at
invoice_items	Invoice lines	invoice_id, sort_order, description, hsn_sac, quantity, unit, unit_price, tax_rate, line_subtotal, discount_amount, taxable_amount, tax_amount, line_total
document_counters	Sequence per business, kind and FY	business_id, kind (invoice/receipt), financial_year, last_number (composite primary key)
payments	Payments and receipts	business_id, invoice_id, receipt_number, amount, amount_inr, payment_date, method, reference, notes
email_logs	Email delivery log	business_id, invoice_id, to_email, subject, status (sent/failed), provider, provider_message_id, error
exchange_rates	Shared rate cache	currency, rate_to_inr, source, fetched_at

5.2 Enumerations

Enum	Values
client_type	b2b, b2c
invoice_status	draft, pending, partially_paid, paid, overdue, cancelled
discount_type	none, percent, fixed
exchange_rate_source	api, manual, base
counter_kind	invoice, receipt
payment_method	bank_transfer, upi, card, cash, cheque, other

Enum	Values
email_status	sent, failed

5.3 Relationships and integrity

- users, clients, invoices, payments, email_logs and document_counters reference businesses and cascade on delete.
- invoices reference clients with **restrict**, so a client with invoices cannot be hard-deleted; archiving is the supported path.
- invoice_items and payments cascade when their invoice is deleted (only drafts can be deleted, and drafts have no payments).
- Indexes exist on business id, status, issue date, client, payment date and the email/invoice relation to keep list and report queries fast.
- The database currently has a single migration that creates 7 enums, 10 tables and 13 indexes.

6. Pages and endpoints

Path	Access	Description
/	Public	Redirects to /dashboard when signed in, otherwise to /login
/login	Public	Sign in; honours a safe ?next= return path
/register	Public	Create business and first user
/dashboard	Signed in	Overview cards, revenue chart, status breakdown, recent activity
/dashboard/clients	Signed in	Client list with search and filters
/dashboard/clients/new, /[id], /[id]/edit	Signed in	Create, view (with invoice history) and edit a client
/dashboard/invoices	Signed in	Paginated invoice list with filters
/dashboard/invoices/new, /[id], /[id]/edit	Signed in	Create, view (with actions, payments, email log) and edit an invoice
/dashboard/payments	Signed in	All payments with filters and INR total
/dashboard/history	Signed in	Merged invoice and payment timeline
/dashboard/records	Signed in	All records: every client, invoice and payment in one tabbed view
/dashboard/reports	Signed in	Date-range reports
/dashboard/settings	Signed in	Business settings
GET /api/invoices/[id]/pdf?download=1	Signed in	Invoice PDF, inline or attachment; 401 if signed out, 404 if not owned
GET /api/payments/[id]/receipt?download=1	Signed in	Receipt PDF, same behaviour

All data changes go through server actions rather than public JSON endpoints: register, login, logout; create/update/archive/restore client; create/update/issue/cancel/delete-draft invoice and send invoice email; record/delete payment; update settings; type-ahead search.

7. User interface and design system

- **Layout:** collapsible icon sidebar, sticky header with breadcrumb, global search, notifications and account menu, and a responsive content area. Tables hide secondary columns on small screens; the sidebar becomes a sheet on mobile.
- **Design language:** soft neumorphic surfaces. Base radius 14px (12 to 16px range) and a dual box-shadow token (light highlight top-left, dark shadow bottom-right) exposed as the shadow-neu utility, used on the sign-in card and the active settings section.
- **Theming:** seven accent palettes override only the accent roles (primary, ring, sidebar accent, chart colours) on top of a neutral base, each with light and dark variants. Mode and theme are stored in localStorage and applied by an inline script before hydration.
- **Components:** shadcn/ui on Base UI primitives for dialogs, dropdowns, selects, popover calendars, tabs, switches, tooltips and tables; charts are pure CSS, so no charting dependency is needed.
- **Feedback:** toast notifications for every action, inline field errors from zod, skeleton loading states and an error boundary with database-connection guidance.

- **Status colours:** draft grey, pending amber, partially paid blue, paid green, overdue red, cancelled struck through.

8. Security

- **Passwords** are hashed with Node's built-in `scrypt` (N=16384, r=8, p=1, 64-byte key, 16-byte random salt) and compared in constant time. The stored format records the parameters so they can be raised later.
- **Sessions:** a 256-bit random token is set in an `httpOnly`, `SameSite=Lax` cookie (Secure in production) for 30 days. The database stores only the SHA-256 hash of the token, so a leaked database cannot be used to hijack sessions. Expired sessions are removed on lookup.
- **Authorisation:** every query is scoped to the signed-in user's business; PDF routes return 404 for documents that belong to another business.
- **Input validation** on client and server with shared zod schemas; GSTIN, emails, amounts, dates and rates are strictly pattern-checked; the login redirect target is validated to prevent open redirects.
- **Secrets** live only in environment variables. The repository ignores all `.env` files except the example.
- **Concurrency:** document numbers are allocated atomically and payments lock the invoice row, preventing duplicate numbers and over-payment under parallel use.

9. Configuration, setup and operation

9.1 Environment variables

Variable	Required	Purpose
DATABASE_URL	Yes	PostgreSQL connection string, e.g. postgresql://USER:PASSWORD@HOST:PORT/DATABASE
NEXT_PUBLIC_APP_URL	Recommended	Public base URL used for the View invoice link in emails (no trailing slash)
RESEND_API_KEY	No	Enables real email delivery through Resend. When empty, emails are printed to the server console.
EMAIL_FROM	No	Sender identity, e.g. "Acme Billing <billing@yourdomain.com>". Defaults to the Resend onboarding address.
EXCHANGE_RATE_CACHE_HOURS	No	How long fetched exchange rates are reused. Default 12.

9.2 Local setup

```
npm install
copy .env.example .env.local          # then set DATABASE_URL (and optional keys)
npm run db:migrate                    # applies db/migrations to the database
npm run dev                          # http://localhost:3000
# open /register to create the first business and user
```

9.3 npm scripts

Script	Command	Use
dev	next dev	Development server with hot reload
build	next build	Production build with type checking
start	next start	Serve the production build
lint	eslint	Lint the codebase
db:generate	drizzle-kit generate	Generate a SQL migration from schema changes
db:migrate	drizzle-kit migrate	Apply pending migrations
db:push	drizzle-kit push	Push the schema directly (development only)
db:studio	drizzle-kit studio	Browse the database in Drizzle Studio

9.4 Production notes

- Deploy on any Node.js host. PDF rendering runs in Node (the renderer is marked as a server external package).
- Use a managed PostgreSQL such as Supabase, Neon or RDS; the driver is configured for transaction-mode poolers.
- Set NEXT_PUBLIC_APP_URL to the public domain, configure RESEND_API_KEY and a verified EMAIL_FROM sender.
- Cookies are marked Secure automatically when NODE_ENV is production, so serve over HTTPS.

- Overdue status is recalculated on page load; a scheduled job is not required but can be added later if nightly reminders are introduced.

10. Typical workflows

10.1 First-time setup

- Register with your business name, your name, email and password.
- Complete Business settings: address and state, GST switch and GSTIN, invoice prefix, default currency, payment terms, bank details, default notes and terms.
- Add your clients with their GSTIN, state and preferred currency.

10.2 Billing a client

- Invoices, New invoice. Pick the client, confirm dates and currency, add line items with HSN/SAC and tax rate, apply any discount, review the live totals and click Issue invoice.
- Open the invoice, Preview PDF to check it, then Send email. The client receives the PDF; the send is logged.
- When money arrives, Record payment with the amount, date, method and reference. Download or send the receipt. The invoice moves to partially paid or paid automatically.

10.3 Month and year end

- Reports: pick the financial year or custom range to see invoiced, collected, outstanding, overdue and tax billed, plus monthly trends, top clients and currency exposure.
- History: review the complete timeline of invoices and payments, filtered by client or period.
- Invoices list: filter by Overdue to follow up on late payers.

11. Current limitations and possible extensions

The following are not part of the current release and are natural next steps:

- Multiple users per business, roles and invitations; password reset and email verification.
- Recurring invoices, quotes/estimates and credit notes.
- Payment reminders, receipt emails and a public (no-login) invoice view and pay link.
- Product or service catalogue with saved prices and HSN/SAC codes; logo upload for PDFs.
- CSV/Excel exports, a persisted audit log of user actions, and automated tests.
- Additional currencies beyond INR, USD, EUR, GBP and AED, and rounding to the nearest rupee.

12. Glossary

Term	Meaning
GST / GSTIN	Goods and Services Tax; the 15-character registration number of a taxpayer
CGST / SGST / IGST	Central and State GST charged on intra-state supplies; Integrated GST charged on inter-state supplies
HSN / SAC	Harmonised System of Nomenclature (goods) and Services Accounting Code (services) printed on tax invoices

Term	Meaning
Place of supply	The state whose GST applies; decides whether CGST+SGST or IGST is charged
B2B / B2C	Business-to-business (client has a GSTIN) and business-to-consumer invoices
Financial year (FY)	1 April to 31 March, written 2026-27
Exchange rate (frozen)	INR per one unit of the invoice currency, fixed at invoice creation
Receipt	The numbered document issued for each payment received
Workspace / tenant	One business and all of its data, isolated from other businesses